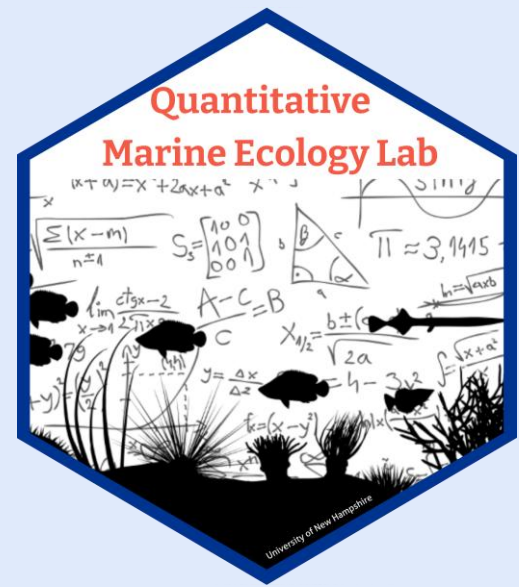


Predicting Intertidal Oyster Thermal Stress with Dynamic Tolerance Models

Andrew Villeneuve^{1*}, Brittany Jellison¹, Easton White¹

¹Department of Biological Sciences, University of New Hampshire. *drew.villeneuve@unh.edu



Introduction

Heatwaves pose increasingly serious risks to marine ectotherms, such as the 2021 PNW heat dome that impacted billions of economically and ecologically important shellfish^{1,2}. Intertidal organisms are adapted to persist in stressful environmental conditions, but thermal stress can be lethal in severe scenarios³. Eastern oysters (*Crassostrea virginica*) are found intertidally in New England, potentially exposing oysters to extreme heat⁴. Predicting the impacts of heatwaves on *C. virginica* requires understanding the role of intertidal exposure and time-dependent thermal tolerance.

Research Questions:

- 1) Does exposure type drive differential oyster thermal tolerance time-dependence?
- 2) Do dynamic tolerance models predict mortality in the field?

Methods

1

Expose adult oysters to static lethal temperature stress (air and water), record time to death using heart rate monitors.

2

Fit Thermal Death Time (TDT) curve and interpolate corresponding Thermal Tolerance Landscape (TTL) for each exposure (Rezende et al. 2014, 2020).

$$\log_{10}(t) = \frac{(CT_{max(1min)} - z)}{T_{assay}}$$

Exposed Submerged

3

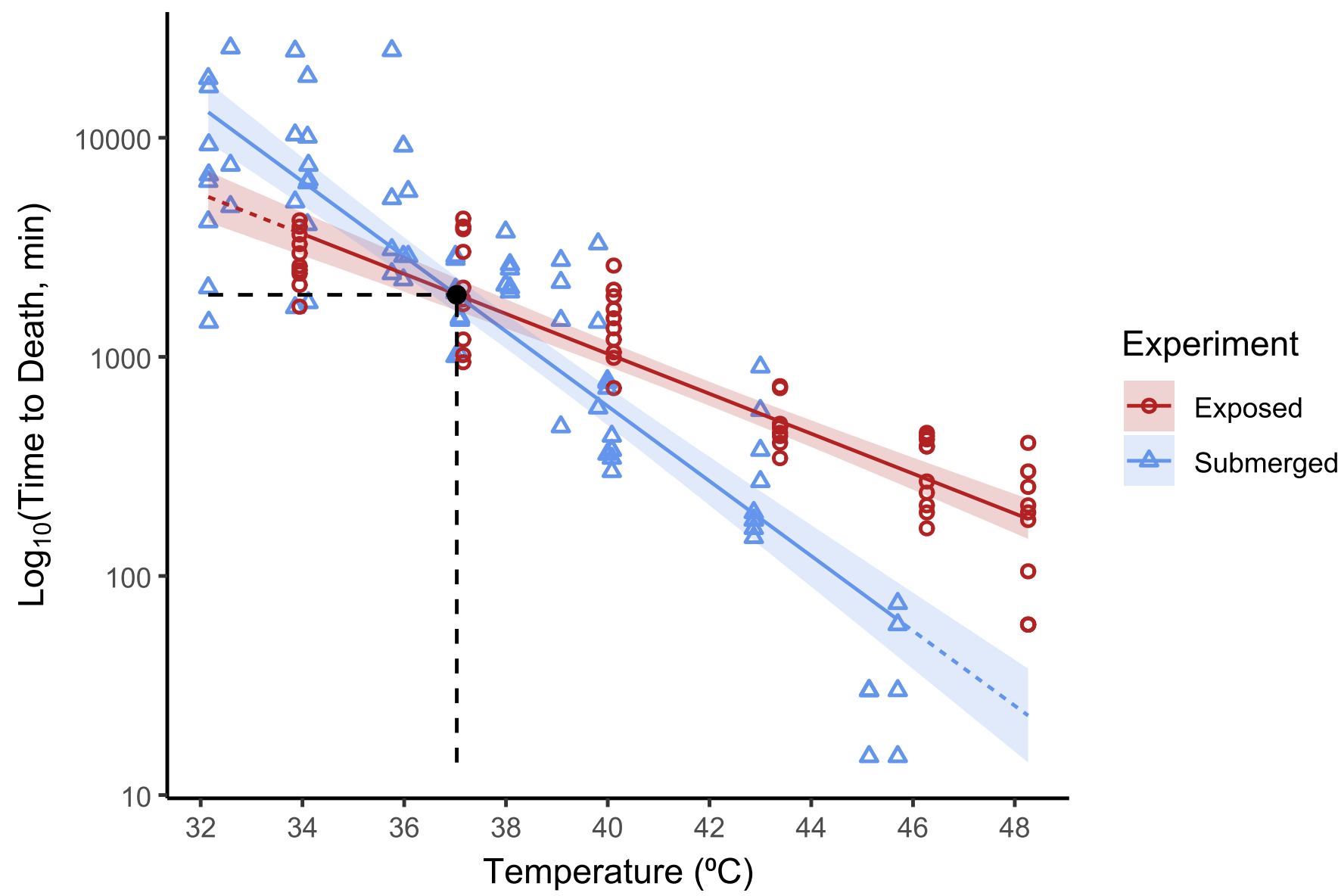
Collect intertidal temperatures and oyster survival ($n = 83$) at two sites in Great Bay, NH during summer 2025 to validate model approach with field data.

4

Intertidal Dynamic Tolerance Algorithm

Dual Thermal Tolerance Landscape Dynamic Model

Results

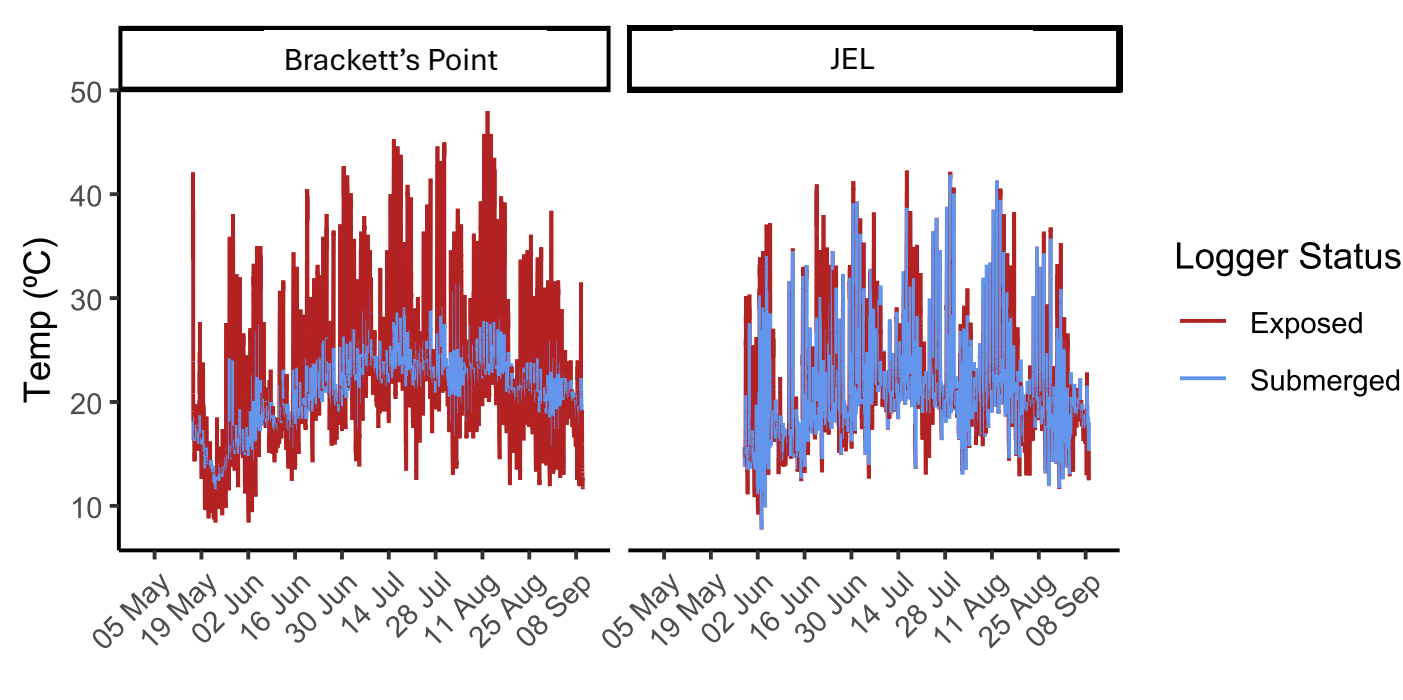


TDT CURVES VARY WITH TIDAL EXPOSURE

The slope of TDT curves for exposed and submerged oysters were significantly different (immersed $\beta = -0.171$, exposed $\beta = -0.091$ $P < 0.001$), suggesting **time dependence of thermal mortality is modulated by tidal exposure**. Heat death occurs **184% faster** during air exposure than in water. TDT curves intersect – below 37°C, submerged oysters persist for longer than exposed oysters, and vice versa above 37°C.

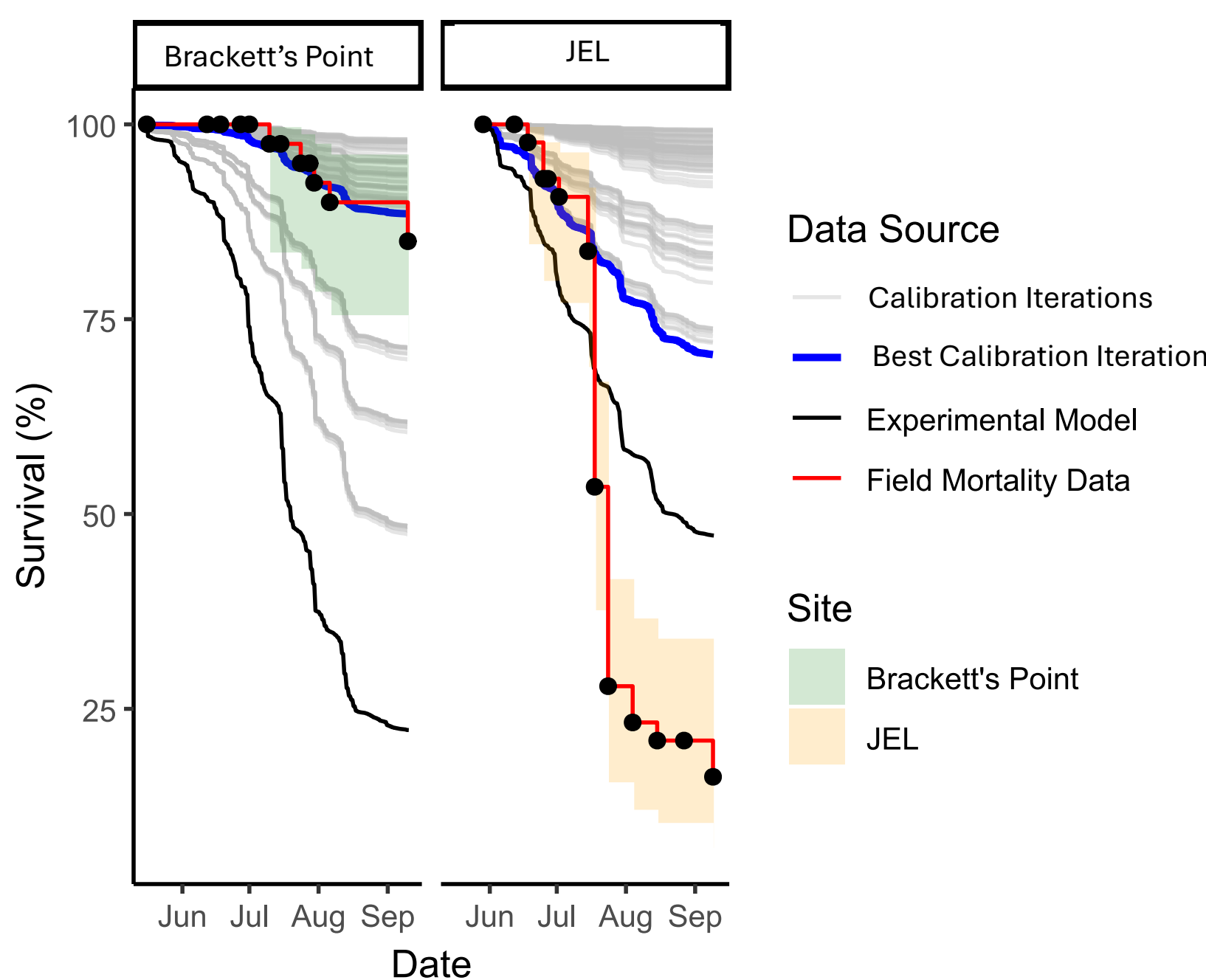
INTERTIDAL MICROCLIMATE

Intertidal temperatures at Brackett's and JEL achieved maximum values of 48.0°C and 42.3°C, respectively.



MODEL VALIDATION

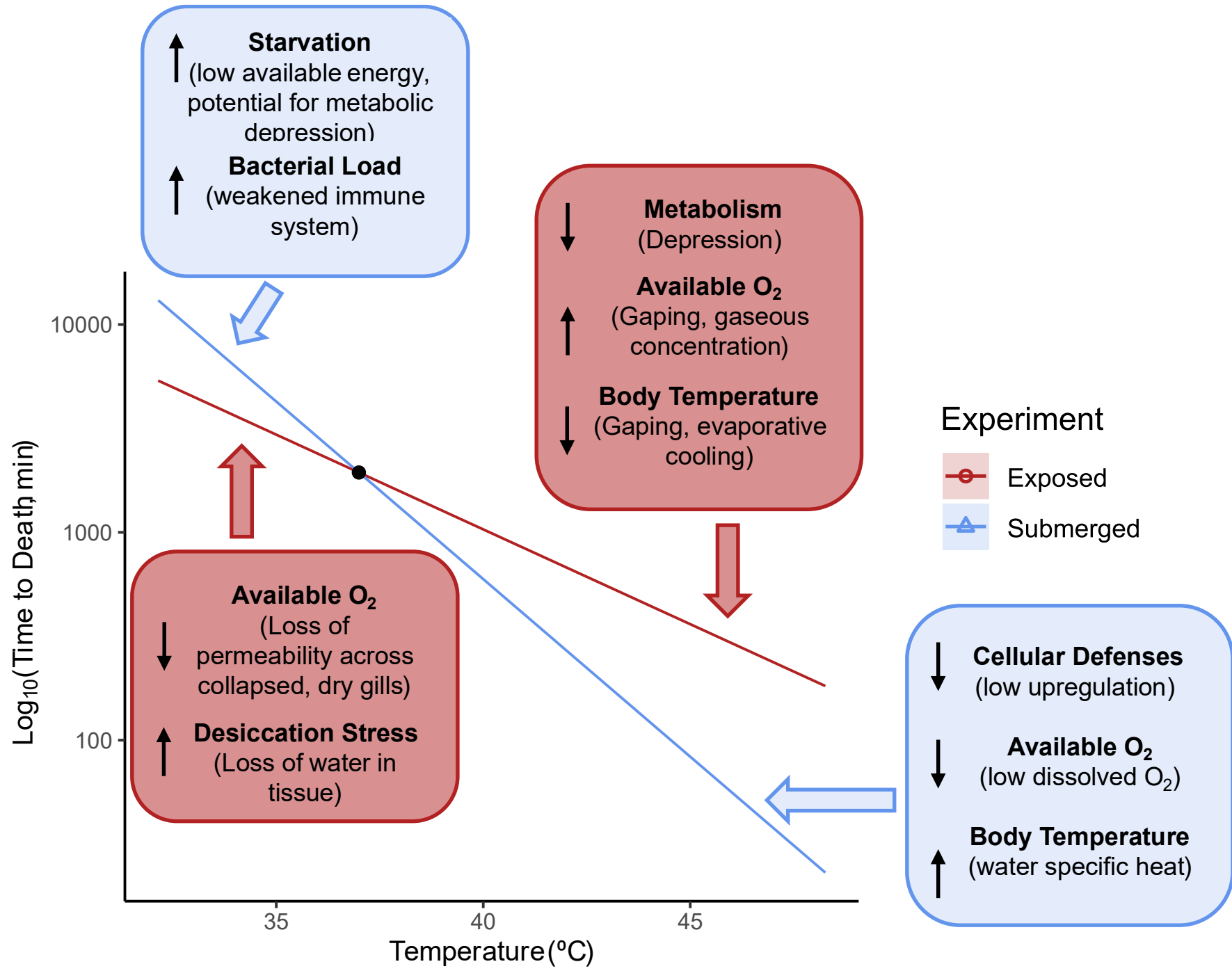
Observed field oyster mortality: Oyster mortality differed between two sites in Great Bay – 15% Brackett's Point, 83.7% JEL. JEL experienced a sudden, steep population decline uncorrelated with temperature in mid-summer.



Dynamic Model Predictions: Dynamic model survival prediction generally overpredicted survival (RMSE Brackett's = 6.4, RMSE JEL = 10.37).

Calibration-based Simulation: Model parameter calibration enhanced model fit (RMSE Brackett's = 0.93, RMSE JEL = 2.29), correcting for static oyster acclimation in experiments vs. variable oyster acclimation in field trials.

Takeaways



- 1 TDT curves change by tidal exposure
Multiple physiological mechanisms likely operate at different time scales to drive mortality. Metabolic depression during tidal emersion likely drives longer persistence at a given temperature than in water (Hui et al. 2022).
- 2 Dynamic tolerance models must consider acclimation plasticity
Our dynamic tolerance model overpredicted survival, likely due to static acclimation of laboratory oysters used to parameterize the model. Field oysters acclimated to variable tidal conditions prior to heat stress, thereby enhancing survival (Morash et al. 2018). Re-calibrating the model to field data provided acceptable model fits.
We were unable to explain the sudden mortality event at JEL with our thermal stress model – other stressors (e.g. *Vibrio*, eutrophication) may be responsible.

Acknowledgements: We thank Josie Demerit, Ella Williams, and Tianyi Duan for their assistance completing laboratory and field experiments. Field work and organism collection was completed under New Hampshire Fish and Game Department permit MFD 2430. This research was supported through grants to ARV by the New Hampshire Sea Grant Graduate Fellowship, and UNH School for Marine Science and Ocean Engineering (SMSOE) Graduate Student Fund. Additional funds were awarded by the New Hampshire Agricultural Experiment Station CREATE program through a joint funding program with the USDA National Institute of Food and Agriculture (Hatch award no. 7004018) and the State of New Hampshire.

Works Cited: 1. Raymond, W. W. et al. Assessment of the impacts of an unprecedented heatwave on intertidal shellfish of the Salish Sea. *Ecology* **103**, e3798 (2022).
2. Smith, K. E. et al. Biological Impacts of Marine Heatwaves. *Annu. Rev. Mar. Sci.* **15**, annurev-marine-032122-121437 (2023).
3. Stillman, J. H. et al. Ecophysiological responses to heat waves in the marine intertidal zone. *Journal of Experimental Biology* **228**, JEB246503 (2025).
4. Heo, J.-M. et al. Impact of exposure temperature rise on mass mortality of tidal flat pacific oysters. *Front. Mar. Sci.* **10**, (2023).
5. Rezende, E. L., Castañeda, L. E. & Santos, M. Tolerance landscapes in thermal ecology. *Funct. Ecol.* **28**, 799–809 (2014).
6. Rezende, E. L., Bozinovic, F., Szilágyi, A. & Santos, M. Predicting temperature mortality and selection in natural *Drosophila* populations. *Science* **369**, 1242–1245 (2020).
7. Hui, T. Y. et al. Timing Metabolic Depression: Predicting Thermal Stress in Extreme Intertidal Environments. *The American Naturalist* **196**, 501–511 (2020).
8. Morash, A. J., Neufeld, C., McCormack, T. J. & Currie, S. The importance of incorporating natural thermal variation when evaluating physiological performance in wild species. *Journal of Experimental Biology* **221**, jeb164673 (2018).